

STRUCTURE THERAPEUTICS (GPCR) — DEEP BIOLOGY + CLINICAL DILIGENCE

As of 2026-09-03 · Biology bench report, filed verbatim to the record at adjudication.

Sourcing: SEC EDGAR, ClinicalTrials.gov API v2, openFDA/Drugs@FDA, PubMed/PMC, Nasdaq short-interest API, live IBKR. Every load-bearing number carries its source. Items the delegated competitive sweep owed are marked NOT-RESEARCHED (see companion GPCR_COMPETITIVE_FIELD_20260903).

Adjudicator's note (2026-09-04): §3's claim that oral semaglutide holds no obesity approval was OVERRULED at adjudication — this bench checked legacy NDA 213051; the competitive bench pulled NDA 218316 directly (WEGOVY tablets, approved 2025-12-22, weight management + MACE indications), and the red bench independently agrees. Two approved orals exist. The remainder of this report stands as filed.

0. THE ONE THING THAT REFRAMES EVERYTHING

Orforglipron is approved and on the market. Not filed — approved.

- **VERIFIED** — openFDA Drugs@FDA, NDA 220934, sponsor Eli Lilly: brand **FOUNDAYO**, ORIG-1 approved **2026-04-01**, review priority STANDARD. Labeling supplement approved 2026-08-04. Six marketed strengths: 0.8 / 2.5 / 5.5 / 9 / 14.5 / 17.2 mg.
- **VERIFIED** — FDA label §2.1: “Take FOUNDAYO orally once daily, **with or without food**.” Titration ≥ 30 days per step, max 17.2 mg.
- Label “Trial 1” = NCT05869903 (ATTAIN-1), n=3,127, results posted 2026-08-14; “Trial 2” = NCT05872620 (ATTAIN-2), n=1,613. Registry and label agree.

Aleniglipron's Phase 3 started 2026-07-13, primary completion estimated **September 2028** — roughly four years behind an approved competitor in the same molecular class and route.

1. PIPELINE STATE

All nine registered studies are sponsored by “Gasherbrum Bio, Inc., a wholly owned subsidiary of Structure Therapeutics” — entity resolution matters; ticker-name search misses them.

NCT	Role	Phase	N	Status	Key dates
NCT06693843	ACCESS (core Ph2b + OLE)	2b	230	Completed	PCD 2025-10-24
NCT06703021	ACCESS II	2	85	Completed	PCD 2026-01-12
NCT07169942	Body Composition	2	71	Active	PCD 2026-08

NCT	Role	Phase	N	Status	Key dates
	(DXA)				
NCT07400588	T2DM + obesity	2	58	Active	PCD 2026-09
NCT07654361	ACCOMPLISH-1	3	3,600 est.	Recruiting	Start 2026-07-13, PCD 2028-09
NCT07654374	ACCOMPLISH-2 (T2DM)	3	1,100 est.	Recruiting	Start 2026-07-17, PCD 2028-09
NCT06139055 / NCT05893043 / NCT05762471	Ph1/1b-2a legacy	1–2a	70/18/142	Completed	2023–24

Extraction-artifact guard — CLEAN. Pivotal NCTs appear verbatim in the company’s own PR; ACCESS in the Nature Medicine paper; remainder resolved from registry by sponsor.

ACCESS (N=230, 36wk) — the properly-powered study

Source: 8-K 2025-12-08 Ex-99.1; Rosenstock et al., Nature Medicine 32:2941–2947 (2026). 5 mg start, titration to 45/90/120 mg. Placebo-adjusted: **–8.2 / –9.8 / –11.3%** (all $p < 0.0001$); placebo –0.8%. At 120 mg: 86% $\geq 5\%$, 70% $\geq 10\%$. SBP –6.4 to –7.5; HR +2.0 to +3.8; no QTc.

Estimand (load-bearing, PR’s own footnote): on-treatment/hypothetical — “efficacy had all randomized participants remained on study treatment... without initiating rescue” — NOT ITT treatment-policy.

Tolerability (Nature Med Table 2): GI TEAE 79–89%; nausea 65–71%; vomiting 32–45%; **AE discontinuation 13.3 / 7.7 / 11.1% (pooled 10.4% = 18/173, verified)** vs placebo 5.4%.

Hepatic — the danuglipron question: company line “no DILI, no persistent liver enzyme elevations, no QTc” is TRUE AS WRITTEN; Extended Data Table 6 shows **$\geq 3\times$ ULN in 8 participants, $\geq 5\times$ ULN in 1, zero $\geq 10\times$** , all resolving on drug. Fully disclosed in the publication, never in a press release — hierarchical disclosure, not concealment; register as absence-of-DILI at $n > 625$, not absence-of-transaminase-signal.

Publication correction logged: Nat Med 32:3095 (2026-06-24): Fig 3b 120mg LS mean –12.1 → **–12.7**; Fig 4 (GI tolerability) initially missing bottom-right panel data and vomiting-panel arrows. Figure-integrity note, not masking.

ACCESS II (N=85) — where the headline comes from

Registered primaries are **safety/PK, not weight**. 44-week: placebo-adjusted **–16.3% (180mg) / –16.0% (240mg)**; **placebo GAINED +1.1%** (vs –0.8% in the powered study — ~2 points of the headline is placebo divergence). **Per-arm N never disclosed anywhere** (~20/arm inferred, not verified; hasResults: false). The superiority claim runs off the smaller, softer-endpointed study; the powered study’s –11.3% is the honest anchor.

Tolerability fix (2.5 mg start — the Phase 3 regimen)

Body Comp interim (~20wk median): 3.4% AE discontinuation. OLE interim (~20wk): **2.0%**; prior-active at 120mg: **16.2% (40.5 lbs) at 56 weeks, no plateau**. ACCESS II $\geq 120\text{mg}$ wk28–44: 1 (3.7%) discontinuation. **All interim, immature windows** — see §3 trap.

Phase 3 ACCOMPLISH — one silent change

Arms 45/90/180mg + placebo, 2.5mg start, wk-76 primary, 71 US sites. Dec-25 and Mar-26 releases both said “up to 240 mg”; actual Phase 3 tops at **180 mg** — disclosed plainly in August, but the 240mg upside-optionality framing quietly retired. The two lower Ph3 doses (45/90) delivered –8.2/–9.8% — below orforglipron’s approved top dose.

Amylin + registry gap

ACCG-2671 (oral DACRA) Ph1 SAD ongoing incl. **food effect**, data + MAD guided Q3-26; ACCG-3535 Ph1 Q4-26; combo (NHP additive data, ADA 2026) initiation Q4-26. Declared: GIPR, GCGR, apelin/APJ, LPA1R programs. **SWITCH trial: guided Q4-26 data, NO registry record under any resolvable term — existence UNVERIFIABLE; do not underwrite that catalyst.**

2. THE BIOLOGY, MECHANISTICALLY

Company claim: aleniglipron “designed to be a biased GPCR agonist, selectively activating G-protein signaling.” **Verification: UNVERIFIABLE — zero PubMed hits for GSBR-1290;** no published cAMP/ β -arrestin bias factor, residence time, or binding data. Company assertion only.

- **Why bias is sought:** GLP-1R couples Gs (cAMP $\rightarrow$ satiety/insulinotropy) and β -arrestin (internalization/desensitization). The design thesis: less arrestin = sustained surface receptor + possibly decoupled nausea. **ACCESS does not support the decoupling** — nausea 65–71%, vomiting 32–45% are at/above class high end.
- **Efficacy ceiling:** observed ~–12% absolute at 36wk (powered), ~–15/–16% at 44–56wk in small/open-label settings, no plateau. Orals cluster low-to-mid teens; injectable tirzepatide/retatrutide a tier above. Ceiling not yet demonstrably broken.
- **Hepatotoxicity — molecule-specific, not class:** danuglipron (2x/day) died on ALT; TERN-601 died on adjudicated DILI; **FOUNDAYO’s label carries no hepatic warning in §5 at all;** aleniglipron clean at $n > 625$. Two structurally distinct molecules cleared the bar danuglipron failed. Class hypothesis not supported.
- **Food effect — settled competitive fact, open GPCR question:** FOUNDAYO “with or without food”; peptide orals require fasting + water limits. Aleniglipron as a small molecule should land in the FOUNDAYO camp — **but no aleniglipron food-effect data has ever been disclosed,** and the Ph3 registry text doesn’t specify. The ACCG-2671 Ph1 explicitly includes food effect; the lead program never reported one. Cheap management question.
- **What amylin adds:** hindbrain/area-postrema satiation on a partly distinct pathway; hoped additive loss, milder aversion, lean-mass preservation. **None verified for ACCG-2671 — no**

human data. Body Comp study (lean mass secondary, Q4-26) is the first real lean datum, and it's for aleniglipron alone.

3. COMPETITIVE COMPARISON — BOTH SIDES FROM PRIMARY

	Aleniglipron ACCESS 120mg	FOUNDAYO 17.2mg
N (arm)	63	730
Duration	36 wk	72 wk
Estimand	on-treatment/hypothetical	ITT, imputation
Absolute	–12.1 (LS –12.7)	–11.1
Placebo-adj	–11.3	–9.0
Placebo arm	–0.8%	–2.1%
≥10%	70%	54.5%
Nausea	65.1%	35%
Vomiting	31.7%	24%
AE discount	11.1% (10.4% pooled)	10% (8% pooled)
Hepatic warning	n/a	none

THE ESTIMAND TRAP (sharpest finding): –11.3% on-treatment vs –9.0% ITT-with-imputation are non-comparable quantities; the on-treatment estimand systematically flatters. The company's cross-class comparisons never flag this. ACCOMPLISH will be scored on regulatory ITT → **Phase 3 headline should print below Phase 2 for estimand reasons alone.**

THE TOLERABILITY-WINDOW TRAP: 2.0–3.4% at ~20wk median interim vs completed 72-week label rates. Matched basis (5mg/36wk): aleniglipron 10.4% sits ON TOP of FOUNDAYO's 72-week 8–10%, nausea ~2x, GI +10 points. The differentiation claim has no primary support yet.

External anchor: Kamrul-Hasan et al. NMA (Obes Sci Pract 2026;12(4):e70181, 9 RCTs, N=5,766): oral semaglutide **–11.6%**, orforglipron **–11.8%** placebo-adjusted. ACCESS's –11.3% is squarely in the pack. Only the N=85 study separates aleniglipron.

M&A demand side (verified in full): Pfizer/Metsera — merger 2025-09-21, amended 11-07 (contested-auction fingerprint), closed 11-13: \$65.60 + CVR ≤\$20.65 (max \$86.25). Pfizer killed danuglipron on liver, paid ~\$10B for a Phase-2 INJECTABLE portfolio, still holds no clinical oral. **Structure hired Metsera's CLO through that deal (Matthew Lang) as COO/GC eff. 2026-04-15, and Novo's obesity commercial exec (John Berrios) as CCO eff. ~2026-08-06** — a company preparing both paths, foreclosing neither.

4. FINANCIAL STATE (10-Q 6/30/26, acc. 0001104659-26-092057)

Cash+ST **\$1,342.7M**; liabilities \$71.0M; **no debt, no preferred**; accumulated deficit \$652.4M. Q2 R&D \$100.1M (+83% YoY); Q2 net loss \$106.2M; **Q2 standalone operating burn ~\$120M** (H1 – \$104.9M flattered by the January Genentech \$100M). Runway “through end of 2028... **excluding pre-commercialization**” = \$134M/q average — arithmetically coherent, no cushion, commercial build carved out. **Additional equity before 2028 data is the base case.**

Cap structure: 213,655,000 ordinary (71.22M ADS) + pre-funded warrants 4,615,386 ordinary (1.54M ADS, \$0.0001, perpetual, unexercised) → **fully-inclusive 72.76M ADS**. Dec-25 follow-on: 9.96M ADS @ \$65.00 + warrants, net \$701.5M, priced 3 days post-topline — buyers –27%. ATM amended May-26: \$400M registered, ~\$341.5M available, cap removed from agreement; S-3ASR shelf.

Genentech/Roche license (bigger than \$100M): non-exclusive sublicensable PATENT license + covenant, covering CT-996; \$100M upfront (received Jan-26) + low-single-digit royalties. Risk factor states plainly: cannot block CT-996, but collects royalties — **arm's-length validation of blocking strength over a competing oral chemotype, and an inversely-hedged royalty stream**. IP: 24 GLP-1R families to 2041–2045; amylin 13 families (Schrödinger milestones ≤\$89M); aleniglipron royalty-free.

Holders: FMR >10% but trimmed ~9% of position H1-26 (31.1M → 28.2M ordinary); Wellington, Deep Track, Avoro, BVF, Janus. **Insider selling = mechanical 10b5-1** (CSO Lin, fixed cadence, 2019 options at \$0.34) — not a signal. No open-market buying on any Form 4 reviewed.

5. CATALYST MAP (no 8-K since 2026-08-06; all Q3 events pending, 27 days left)

Event	Guided	Status
ACCESS OLE 72-wk topline (incl. 180mg exposure)	Q3-26	pending, no date
ACCG-2671 SAD data (incl. food effect) + MAD start	Q3-26	pending; trial unregistered
Body Comp topline (lean mass secondary)	Q4-26	registry PCD supports
T2DM trial data	Q4-26	registry PCD supports
SWITCH data	Q4-26	UNVERIFIABLE — no registry record
ACCG-3535 init · combo init	Q4-26	company-guided
ACCOMPLISH-1/-2 primary completion	2028-09	registry-confirmed

Catalyst 1 — OLE 72-wk (≈75% by 9/30, 95% mid-Oct): the decisive line is **mature discontinuation at 180mg on the 2.5mg regimen**, not the open-label weight number. Skew ~60/40 benign, negatively convex (–20/–30% if discontinuation converges to 8–10%; +10/20% if low single digits hold). Tripwires: 7.01/8.01 8-K two-exhibit format; leading-with-weight/burying discontinuation = informative framing; 2.5mg-subgroup-only reporting; ACCOMPLISH registry edits around the print; silence past 9/30.

Catalyst 2 — ACCG-2671 SAD (≈70% by 9/30): tests whether a second real asset exists (route to re-rating independent of aleniglipron; no approved oral amylin exists anywhere). Watch food effect, nausea at tolerated exposures, once-daily PK. ~65% benign, +5/15% vs –10/15%. **Tripwire: SAD data WITHOUT simultaneous MAD initiation = the tell.**

Q4 cluster = one correlated event, not four (one leg unverifiable).

6. TAPE (IBKR live, 2026-09-03 close)

\$47.34; 52wk \$18.49–94.90; –50% from high, +2.6x off low; YTD –31.9%; \$35.3M/day 90d dollar volume; **RV-30d 50.7%**. Mcap \$3.444B (fully-inclusive), **EV ≈ \$2.10B**. Short interest 6.08M (≈8.5% of ADS), DTC 10.4, building slowly. Options exist (2028 tenor brackets the pivotal): Oct straddle ≈22.7% implied move, Dec ≈40% — but **ATM open interest in the TENS, spreads 20–55% wide; indicative only, untransactable in size.**

7. HONESTY-OF-PRESENTATION FINDINGS

1. Superiority claim runs off the weaker study (N=85, safety-primary, weight-gaining placebo).
2. Estimand mismatch never flagged while making explicit cross-class comparisons.
3. Tolerability comparison uses an immature window.
4. The hepatic line is precisely drawn (true as written; excursions only in Extended Data).
5. **The Q2 10-Q — filed four months after Foundayo’s approval — contains zero occurrences of “orforglipron”** (nor Rybelsus/Wegovy/Zepbound); the same risk factor WAS updated for Metsera.
6. Anti-masking symmetry: 10.4% reconciles exactly; full GI table, hepatic data, SAE imbalance, HR increase all published; dose reduction disclosed; Genentech adverse element stated; runway carve-out unusually candid. **Discloses bad news in filings/publications, frames optimistically in PRs — materially different from concealment.**

8. NOT FOUND / UNVERIFIABLE (no silent caps)

ACCESS II per-arm N · SWITCH registry record · ACCG-2671/-3535 registry records · aleniglipron biased-agonism pharmacology (zero PubMed) · aleniglipron food-effect data · FDA EOP2 minutes · ACCESS registry results (FDAAA deadline 2026-10-24) · IR events page (cookie-walled) · ACCESS II 44-wk denominators. Competitive items owed by the delegated sweep are in the companion report; the two that could shade probabilities (elecoglipron Ph2 data, CT-996 Ph2 data) are covered by the gap sweep dispatched 2026-09-04.